Spatio-Temporal Analysis of Team Sports

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Team-based *invasion sports* such as football, basketball, and hockey are similar in the sense that the players are able to move freely around the playing area and that player and team performance cannot be fully analysed without considering the movements and interactions of all players as a group. State-of-the-art object tracking systems now produce *spatio-temporal* traces of player trajectories with high definition and high frequency, and this, in turn, has facilitated a variety of research efforts, across many disciplines, to extract insight from the trajectories. We survey recent research efforts that use spatio-temporal data from team sports as input and involve non-trivial computation. This article categorises the research efforts in a coherent framework and identifies a number of open research questions.

CCS Concepts: • **General and reference → Surveys and overviews**; • **Information systems → Geographic information systems**; *Data mining*; *Data analytics*; • **Theory of computation → Theory and algorithms for application domains;**

Additional Key Words and Phrases: Trajectory, spatio-temporal data, sports analysis, spatial subdivision, network analysis, data mining, machine learning, performance metrics, basketball, soccer, football, american football, handball, hockey

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1. INTRODUCTION

Team sports are a significant recreational activity in many societies and attract participants to compete in, watch, and capitalise from the sport. There are several sporting codes that can be classed together as *invasion sports* in that they share a common structure: two teams are competing for possession of a ball (or puck) in a constrained playing area, for a given period of time, and each team has simultaneous objectives of scoring by putting the ball into the opposition's goal, while also defending their goal against attacks by the opposition. The team that has scored the greatest number of goals at the end of the allotted time is the winner. Football (soccer), basketball, ice hockey, field hockey, rugby, Australian Rules football, American football, handball, and Lacrosse are all examples of invasion sports.

Teams looking to improve their chances of winning will naturally seek to understand their performance and that of their opposition. Systematic analysis of sports play is not new—box-scores for summarising baseball matches have existed since 1860

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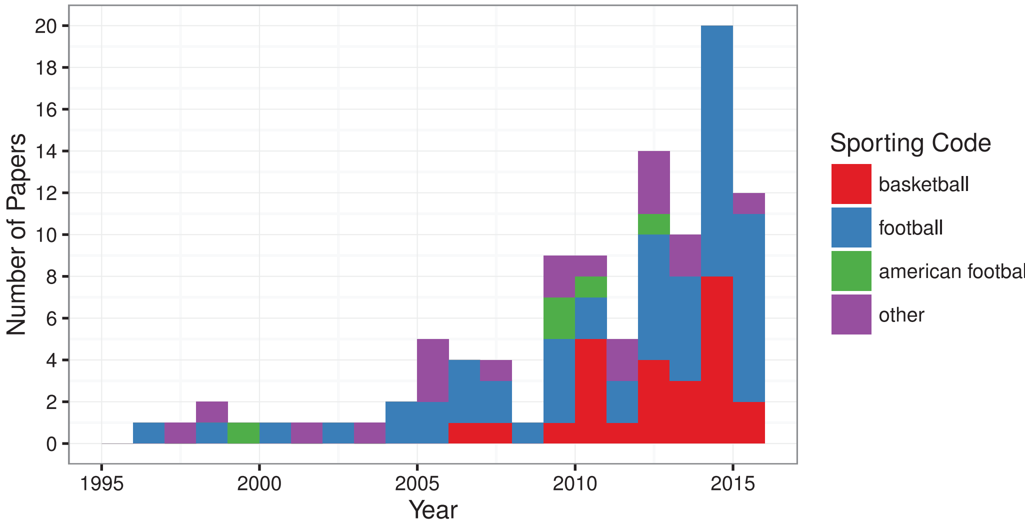


Fig. 1. Spatial sports research articles cited in this survey, by year, 1995–2015 (to date), divided by sporting code. There has been a significant increase in articles published in this area as data have become available for researchers, particularly in football and basketball.

[Lamoreaux 1977], and manual notation of events in football matches has occurred from the 1950s using manual notation methods [Reep and Benjamin 1968]. However, human observation can be unreliable—experimental results in Franks and Miller [1986] showed that the expert observers' recollection of significant match events is as low as 42%—and, in recent years, automated systems to capture and analyse sports play have proliferated.

Today, there are a number of systems in use that capture spatio-temporal data from team sports during matches. The adoption of this technology and the availability to researchers of the resulting data varies among the different sporting codes and is driven by various factors, particularly commercial and technical. There is a cost associated with installing and maintaining such systems, and while some leagues mandate that all stadium have systems fitted, in others the individual teams will bear the cost and thus view the data as commercially sensitive. Furthermore, the nature of some sports present technical challenges to automated systems, for example, sports such as rugby and American football have frequent collisions that can confound optical systems that rely on edge detection.

To date, the majority of datasets available for research are sourced from football and basketball, and the research we surveyed reflects this; see Figure 1. The National Hockey League intends to install a player tracking system for the 2015–2016 season, and this may precipitate research in ice hockey in coming years [SportVision Inc. 2015].

Sports performance is actively researched in a variety of disciplines. To be explicit, the research that we consider in this survey fulfils the following three key criteria:

(1) We consider **team-based invasion sports**.
(2) The model used in the research has **spatio-temporal data** as its primary input.
(3) The model performs some **non-trivial computation** on the spatio-temporal data. In other words, a novel algorithmic approach is presented or applied.

The research covered has come from many communities, including machine learning, network science, Geographic Information Science (GIS), computational geometry, computer vision, complex systems science, statistics, and sports science. There has been a

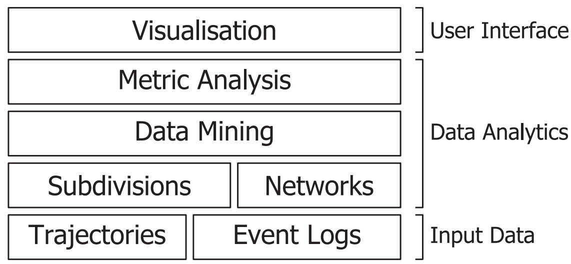


Fig. 2. Summary of the major approaches surveyed, with each approach corresponding to a section of this article. The techniques described in a particular layer of this diagram can be used as input to a technique described in a higher layer.

consequent diversity of methods and models used in the research, and our intention in writing this survey was to provide an overview and framework on the research efforts to date.

Furthermore, the spatio-temporal data extracted from sports has several useful properties that make it convenient for fundamental research. For instance, player trajectories exhibit small spatial and temporal range, dense sampling rates, a small number of agents (i.e., players), highly cooperative and adversarial interaction between agents, and a latent structure. As such, we believe that this survey is a timely contribution to this emerging area of research.

This survey contains the following sections, summarised in Figure 2. In Section 2, we describe the primary types of spatio-temporal data captured from team sports. We describe the properties of these data and outline the sports from which it is currently available.

Section 3 describes approaches that have been used to subdivide the playing area into regions that have a particular property. The playing area may be discretized into a fixed subdivision and the occurrences of some phenomena counted, for instance, a player occupying a particular region or a shot at goal occurring from that region, producing an *intensity map* of the playing area. On the other hand, a subdivision of the playing area based on areas that are dominated by particular players has also been used in several articles.

In Section 4, we survey approaches that represent temporal sequences of events as *networks* and apply network-theoretic measures to them. For example, sequences of passes between players can be represented as a network with players as the vertices and weighted edges for the frequency of passes between pairs of players, and network measures be computed to quantify the passing performance.

Section 5 provides a task-oriented survey of the approaches to uncover information inherent in the spatio-temporal data using *data-mining* techniques. Furthermore, several articles define metrics to measure the performance of players and teams, and these are discussed in Section 6.

We detail the research into *visualisation* techniques to succinctly present metrics of sports performance in Section 7.

Table I summarises the techniques surveyed in this article and indicates the particular sports that they have been applied to. In Section 8, we discuss the factors may be considered when applying these techniques to other sports.

2. REPRESENTING SPORTS PLAY USING SPATIO-TEMPORAL DATA

The research surveyed in this article is based on *spatio-temporal data*, the defining characteristic of which is that it is a sequence of samples containing the time-stamp and location of some phenomena. In the team sports domain, two types of spatio-temporal

Table I. Summary of the Approaches and Techniques Described in this Survey and the Sports in Which They Have Been Applied

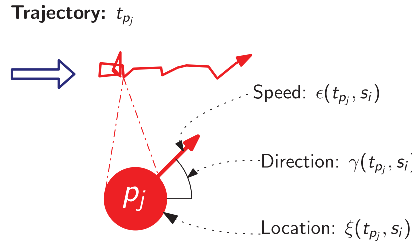
	Football	Basketball	Field Hockey	Ice Hockey	American Football	Handball
3. Playing Area Subdivision						
3.1. Playing Area Subdivision	⋆	⋆				
3.2. Low-Rank Factor Matrices		⋆				
3.3. Movement Models and Dominant Regions	⋆	⋆				
4. Network Techniques for Team Performance Analysis						
4.1. Centrality	⋆	⋆				
4.2. Clustering Coefficients	⋆	⋆				
4.3. Density and Heterogeneity	⋆					
4.4. Entropy, Topological Depth, Price-of-Anarchy, and Power-Law Distributions		⋆				
5. Data Mining						
5.1. Applying Labels to Events	⋆	⋆				
5.2. Predicting Future Event Types and Locations	⋆	⋆				
5.3. Identifying Formations	⋆	⋆	⋆			
5.4. Identifying Plays and Tactical Group Movement	⋆				⋆	
5.5. Temporally Segmenting the Game	⋆	⋆				⋆
6. Performance Metrics						
6.1. Offensive Performance	⋆	⋆				
6.2. Defensive Performance		⋆				
7. Visualisation	⋆	⋆		⋆		

data are commonly captured: *object trajectories* that capture the movement of players or the ball and *event logs* that record the location and time of match events, such as passes, shots at goal, or fouls. These datasets, described in detail below, individually facilitate the spatio-temporal analysis of play; however, they are complementary in that they describe different aspects of play and can provide a richer explanation of the game when used in combination. For example, the spatial formation in which a team arranges itself in will be apparent in the set of player trajectories. However, the particular formation used may depend on whether the team is in possession of the ball, which can be determined from the event log. On the other hand, a *shot at goal* event contains the location from where the shot was made, but this may not be sufficient to make a qualitative rating of the shot. Such a rating ought to consider whether the shooter was closely marked by the defence and the proximity of defenders—properties that can be interpolated from the player trajectories.

2.1. Object Trajectories

The movement of players or the ball around the playing area are sampled as a timestamped sequence of location points in the plane; see Figure 3. The trajectories are captured using optical- or device-tracking and processing systems. *Optical tracking systems* use fixed cameras to capture the player movement, and the images are then processed to compute the trajectories [Bradley et al. 2007]. There are several commercial vendors who supply tracking services to professional sports teams and leagues [ChyronHego Corporation 2015; Impire AG 2015; Prozone Sports Ltd 2015; STATS LLC 2015]. On the other hand, *device tracking systems* rely on devices that infer their

p	s_i	x_1	x_2
...			
p_j	10.1	3940	-2362
p_j	10.2	3948	-2346
p_j	10.3	3956	-2331
p_j	10.4	3923	-2392
p_j	10.5	3977	-2309
...			



s_i	v	P
...		
10.1	*Touch*	$\{p_j\}$
10.2	*Pass*	$\{p_j\}$
10.6	*Touch*	$\{p_k\}$
11.0	*Tackle*	$\{p_l, p_k\}$
...		

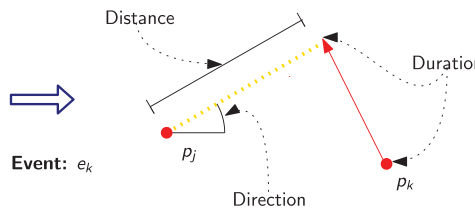


Fig. 3. Example of the trajectory and event input data and an illustration of their geometric representations. Each trajectory is a sequence of location points, and these can be used to extrapolate the basic geometry of a player at a given timestep. Similarly, the geometry of events such as the *pass* shown can be computed from the trajectories of the involved players.

location and are attached to the players' clothing or embedded in the ball or puck. These systems can be based on *Global Positioning System* (GPS) [Catapult Sports Ltd 2015] or *Radio-Frequency Identification* (RFID) [SportVision Inc. 2015] technology.

The trajectories produced by these systems are dense, in the sense that the location points samples are uniform and frequent—in the range of 10 to 30Hz.

The availability of spatio-temporal data for research varies. Some leagues capture data from all matches, such as the National Basketball Association (NBA) [NBA Media Ventures LLC 2014] and the German Football Leagues [Impire AG 2015], in other cases, teams capture data at their stadia only. Leaguewide datasets are not simply larger but also allow for experiments that control for external factors such as weather, injuries to players, and playing at home and on the road.

2.2. Event Logs

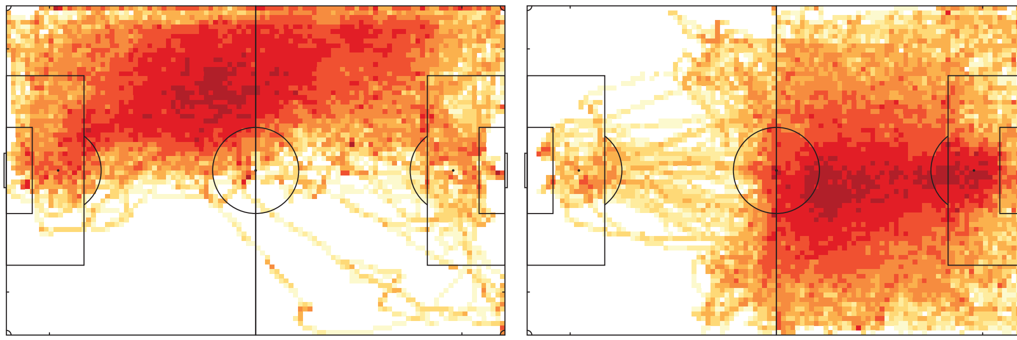
Event logs are a sequence of significant events that occur during a match. Events can be broadly categorised as *player events*, such as passes and shots, and *technical events*, for example, fouls, time-outs, and start/end of period; see Figure 3. Event logs may be derived in part from the trajectories of the players involved; however, they may also be captured directly from video analysis; for example, Opta Sports Ltd [2015] uses this approach. This is often the case in sports where there are practical difficulties in capturing player trajectories, such as in rugby and American football.

Event logs qualitatively differ from the player trajectories in that they are not dense—samples are only captured when an event occurs—however, they can be semantically richer, as they include details like the type of event and the players involved.

The models and techniques described in the following sections all use object trajectories and/or event logs as their primary input.

3. PLAYING AREA SUBDIVISION

Player trajectories and event logs are both low-level representations and can be challenging to work with. One way to deal with this issue is to discretize the playing area into regions and assign the location points contained in the trajectory or event log to a discretized region. The frequency—or intensity—of events occurring in each region



(a) Left-back (b) Striker

Fig. 4. Examples of intensity maps showing areas of the football pitch that the players occupy. The player trajectories have been oriented such that the play is from left to right. (a) The left-back is positioned on the left of the field but is responsible for taking attacking corner-kicks from the right. (b) The striker predominantly stays forward of the half-way line but, however, will retreat to help defend corner-kicks.

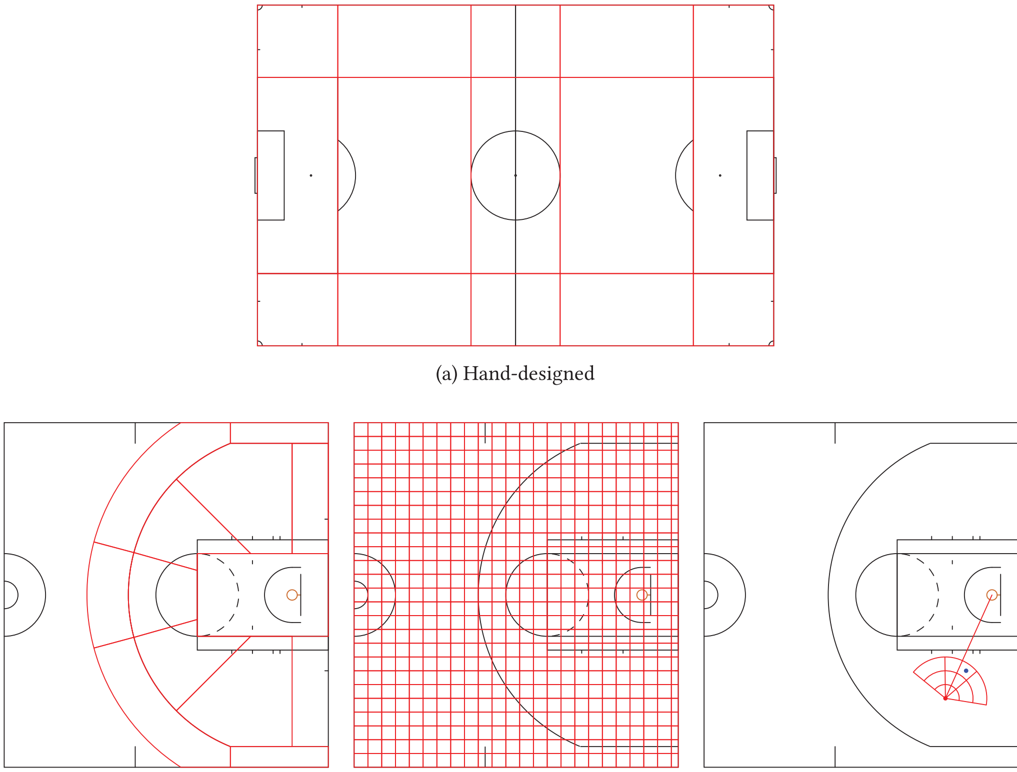
is a spatial summary of the underlying process; alternatively, the playing area may be subdivided into regions such that each region is dominated in some sense by a single player, for example, by the player being able to reach all points in the region before any other player. There are a variety of techniques for producing playing area subdivisions that have been used in the research surveyed here and are summarised in this section.

3.1. Intensity Matrices and Maps

Spatial data from team sports have the useful property that they are constrained to a relatively small and symmetric playing area—the pitch, field, or court. The playing area may be subdivided into regions and events occurring in each region can be counted to produce an intensity matrix and can be visualised with an intensity map; see Figure 4. This is a common preprocessing step for many of the techniques described in subsequent sections.

When designing a spatial discretization, the number and shape of the induced regions can vary. A common approach is to subdivide the playing area into rectangles of equal size [Bialkowski et al. 2014b; Borrie et al. 2002; Cervone et al. 2014; Franks et al. 2015a; Lucey et al. 2012; Narizuka et al. 2014; Shortridge et al. 2014]; for example, see Figure 5(c). However, the behaviour of players may not vary smoothly in some areas. For example, around the three-point line on the basketball court, a player's propensity to shoot varies abruptly, or the willingness of a football defender to attempt a tackle will change depending on whether he or she is inside the penalty box. The playing area may be subdivided to respect such predefined assumptions of the player's behaviour. Camerino et al. [2012] subdivides the football pitch into areas that are aligned with the penalty box; see Figure 5(a), and interactions occurring in each region were counted. Similarly, Maheswaran et al. [2014] and Goldsberry and Weiss [2013] define subdivisions of the basketball half-court that conform with the three-point line and are informed by intuition of shooting behaviour; see Figure 5(b).

Transforming the playing area into polar space and inducing the subdivision in that space is an approach used in several articles. This approach reflects the fact that player behaviour may be similar for locations that are equidistant from the goal or basket. Using the basket as the origin, polar-space subdivisions were used by Reich et al. [2006] and by Maheswaran et al. [2012]. Yue et al. [2014] used a polar-space subdivision to discretize the position of the players marking an attacking player. Under this scheme, the location of the attacking player was used as the origin, and the polar space aligned such that the direction of the basket is at 0°; see Figure 5(d).



(a) Hand-designed

(b) Hand-designed (c) Cartesian grid (d) Polar grid

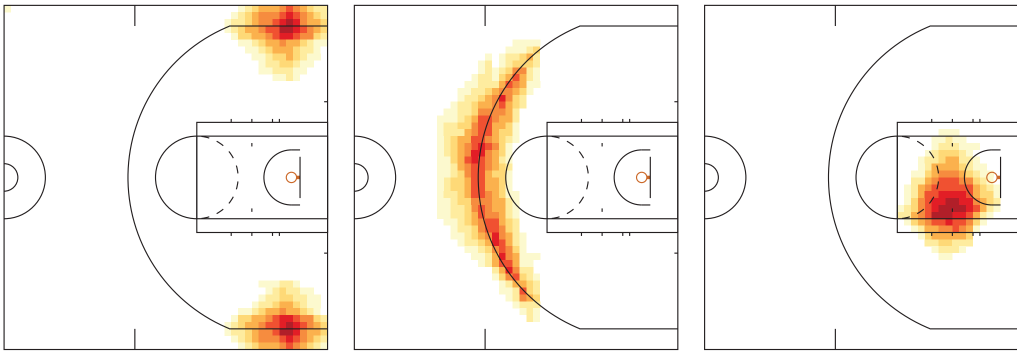
Fig. 5. Examples of subdivisions used to discretize locations: ((a) and (b)) hand-designed subdivision reflecting expert knowledge of game-play in basketball [Maheswaran et al. 2014] and football [Camerino et al. 2012]; (c) subdivision of court into unit-squares [Cervone et al. 2014]; (d) polar subdivision where origin is centred on ball-carrier and grid is aligned with the basket [Yue et al. 2014].

Given a subdivision of the playing area, counting the number of events by each player in each region induces a discrete spatial distribution of players' locations during the match. This can be represented as an $\mathbb{R}_+^{N \times K}$ intensity matrix containing the counts X for N players in each of the R regions of the subdivision. The event X may be the number of visits by a player to the region; for example Maheswaran et al. [2014] used the location points from player trajectories to determine whether a cell was visited. Bialkowski et al. [2014a] used event data such as passes and touches made by football players to determine the regions a player had visited.

The number of passes or shots at goal that occur in each region may also be counted. For example, many articles counted shots made in each region of a subdivision of a basketball court [Franks et al. 2015a; Goldsberry and Weiss 2013; Maheswaran et al. 2012; Reich et al. 2006; Shortridge et al. 2014]. Similarly, Borrie et al. [2002], Camerino et al. [2012], Narizuka et al. [2014], and Cervone et al. [2014] counted the number of passes made in each region of a subdivision of the playing area.

3.2. Low-Rank Factor Matrices

Matrix factorization can be applied to intensity matrices described in Section 3.1 to produce a compact, low-rank representation. This approach has been used in several articles to model shooting behaviour in basketball [Cervone et al. 2014; Franks et al.



(a) Corner three-point (b) Top-of-key three-point (c) Right low-post

Fig. 6. Examples of spatial bases induced by using non-negative matrix factorization. Each basis represents an intensity map of where a subset of players tend to shoot from. Shown are three spatial basis intensity maps that represent defined shooting locations.

2015a; Yue et al. 2014]. The insight that motivates this technique is that similar types of players tend to shoot from similar locations, and so each player's shooting style can be modelled as a combination of a few distinct *types*, where each *type* maps to a coherent area of the court that the players are likely to shoot from.

The input is an intensity matrix $X \in \mathbb{R}^{N \times V}$. Two new matrices $W \in \mathbb{R}_{+}^{N \times K}$ and $B \in \mathbb{R}_{+}^{K \times V}$ are computed such that $WB \approx X$ and $K \ll N, V$. The K spatial bases in B represent areas of similar shooting intensity, and the N players' shooting habits are modelled as a linear combination of the spatial bases. The factorization is computed from X by minimizing some distance measure between X and WB, under the constraint that W and B are non-negative. The non-negativity constraint, along with the choice of distance function encourages sparsity in the learned matrices. This leads to intuitive results: Each spatial basis corresponds to a small number of regions of the half-court, and the shooting style of each player is modelled as the mixture of a small number of bases; see Figure 6 for examples of learned spatial bases.

Miller et al. [2014] used non-negative matrix factorization to represent shooting locations in basketball. They observe that the shooting intensity should vary smoothly over the court space and thus fit a Log-Gaussian Cox Process to infer a smooth intensity surface over the intensity matrix, which is then factorized.

Yue et al. [2014] used non-negative matrix factorization to model several event types: shooting, passing, and receiving. They include a spatial regularization term in the distance function used when computing the matrix factorization and claim that spatial regularization can be seen as a frequentist analog of the Bayesian Log-Gaussian Cox process used by Miller et al. [2014].

Cervone et al. [2014] also used non-negative matrix factorization to find a basis representing player roles, based on their occupancy in areas of the court. Players who are similar to a given player were identified as those who are closest in this basis, and this was used to compute a similarity matrix between players.

3.3. Movement Models and Dominant Regions

A team's ability to control space is considered a key factor in the team's performance and was one of the first research areas in which computational techniques were developed. Intuitively, a player dominates an area if he or she can reach every point in that area before anyone else (see Definition 3.1). An early algorithmic attempt to develop a computational tool for this type of analysis was presented by Taki et al. [1996], which defined the *Minimum Moving Time Pattern*—subsequently renamed the *Motion Model*—and the *Dominant Region*.

3.3.1. Motion Model. The motion model presented by Taki et al. [1996] is simple and intuitive: It is a linear interpolation of the acceleration model. It assumes that potential acceleration is the same in all directions when the player is standing still or moving very slowly. As speed increases, it becomes more difficult to accelerate in the direction of the movement. However, their model did not account for deceleration and hence is only accurate over short distances.

Fujimura and Sugihara [2005] presented a more realistic motion model; in particular, they incorporated a resistive force that decrease the acceleration. The maximum speed of a player is bounded, and, based on this assumption, Fujimura and Sugihara [2005] formulated the following equation of motion:

$$m\frac{d}{dt}v = F - kv, \tag{1}$$

where m is the mass, F is the maximum driving force, k is the resistive coefficient, and v is the velocity. The solution of the equation is as follows:

$$v = \frac{F}{k} - \left(\frac{F}{k} - v_0\right) \cdot \exp\left(-\frac{k}{m}t\right),$$

where v_0 is the velocity at time $t = 0$. If the maximum speed $v_{\max} = F/k$ and the magnitude of the resistance $\alpha = k/m$ are known, then the motion model is fixed. To obtain α and $v_{\max}$, Fujimura and Sugihara [2005] studied players' movement on video and empirically estimated α to be 1.3 and $v_{\max}$ as 7.8m/s. This is then generalised to two dimensions as follows:

$$m\frac{d}{dt}\mathbf{v} = \mathbf{F} - k\mathbf{v}.$$

Solving the equation, we get that all the points reachable by a player, starting at position x_0 with velocity $\mathbf{v_0}$, can reach point x within time t form the circular region centred at

$$x_0 + \frac{1 - e^{-\alpha t}}{\alpha} \cdot \mathbf{v_0} \quad \text{with radius} \quad \mathbf{v}_{\max} \cdot \frac{1 - e^{-\alpha t}}{\alpha}.$$

They compared this model empirically and observed that the model yields a good approximation of actual human movement, but they stated that a detailed analysis is a topic for future research.

A different model was used in a recent article by Cervone et al. [2014], with the aim to predict player movement in basketball. They present what they call a micro-transition model. The micro-transition model describes player movement during a single possession of the ball. Separate models are then used for defense and attack. Let the location of an attacking player ℓ at time t be $(x^\ell(t), y^\ell(t))$. Next they model the movement in the x and y coordinates at time $(t + \varepsilon)$ using

$$x^\ell(t + \varepsilon) = x^\ell(t) + \alpha_x^\ell[x^\ell(t) - x^\ell(t - \varepsilon)] + \eta_x^\ell(t) \tag{2}$$

and analogously for $y^\ell(t + \varepsilon)$. This expression derives from a Taylor series expansion of the function for determining the ball-carrier's position such that $\alpha_x^\ell[x^\ell(t) - x^\ell(t - \varepsilon)] \approx \varepsilon x^\ell(t)$, and $\eta_x^\ell(t)$ represents the contribution of higher-order derivatives modelling accelerations and jerks. When a player receives the ball outside the three-point line, the most common movement is to accelerate towards the basket. On the other hand, a player will decelerate when closer to the basket. Players will also accelerate away from

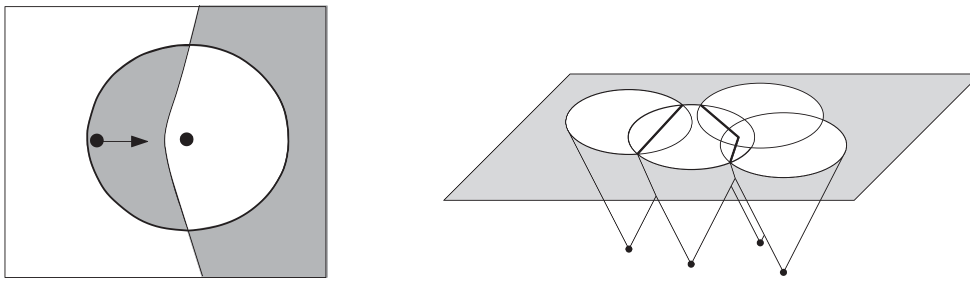


Fig. 7. (a) Showing the dominant region for two players. The left player is moving to the right with high speed and the right player is standing still. Using the motion models discussed in Section 3.3.1, the resulting dominant region for a single player might not be connected. (b) A standard approach used in computational geometry to subdivide the plane is to compute the projection of the intersection of surfaces in three dimensions onto the plane.

the boundary of the court as they approach it. To capture this behaviour, the authors suggest mapping a player's location to the additive term $\eta_x^\ell(t)$ in Equation (2). The positions of the five defenders are easier to model when conditioned on the evolution of the attack's positions; see Cervone et al. [2014] for details.

Next we consider how the motion models have been used to develop other tools.

3.3.2. Dominant Regions. The original article by Taki et al. [1996] defined the dominant region as follows.

Definition 3.1. The *dominant region* of a player p is the region of the playing area where p can arrive before any other player.

Consequently, the subdivision induced by the dominant regions for all players will partition the playing area into cells. In a very simple model where acceleration is not considered, the dominant region is equivalent to the Voronoi region, and the subdivision can be efficiently computed [Fortune 1987]. However, for more elaborate motion models, such as the ones described in Section 3.3.1, the distance function is more complex. For some motion models, the dominant region may not be a connected area [Taki and Hasegawa 2000]; an example is shown in Figure 7(a). A standard approach used to compute the subdivision for a complex distance function is to compute the intersection of surfaces in three dimensions, as shown in Figure 7(b). However, this is a complex task and time consuming for non-trivial motion models. Instead, approximation algorithms have been considered in the literature.

Taki and Hasegawa [1998, 2000] implemented algorithms to compute dominant regions, albeit using a simple motion model. Instead of computing the exact subdivision, they considered the 640×480 pixels that at that time formed a computer screen, and for each pixel they computed the player who could reach that pixel first, hence, visualizing the dominant regions. The same algorithm for computing the dominant region was used by Fujimura and Sugihara [2005], although they used a more realistic motion model; see Section 3.3.1.

However, the above algorithms were shown to be slow in practice; for example, preliminary experiments by Nakanishi et al. [2009] stated that the computation requires 10 to 40s for a 610×420 grid. To achieve the real-time computation required for application in the RoboCup robot football tournament [Kitano et al. 1997], the authors proposed an alternative approach. Instead of computing the time required for every player to get to every point, Nakanishi et al. [2009] used a so-called *reachable polygonal region* (RPR). The RPR of a player p given time t is the region that p can reach within

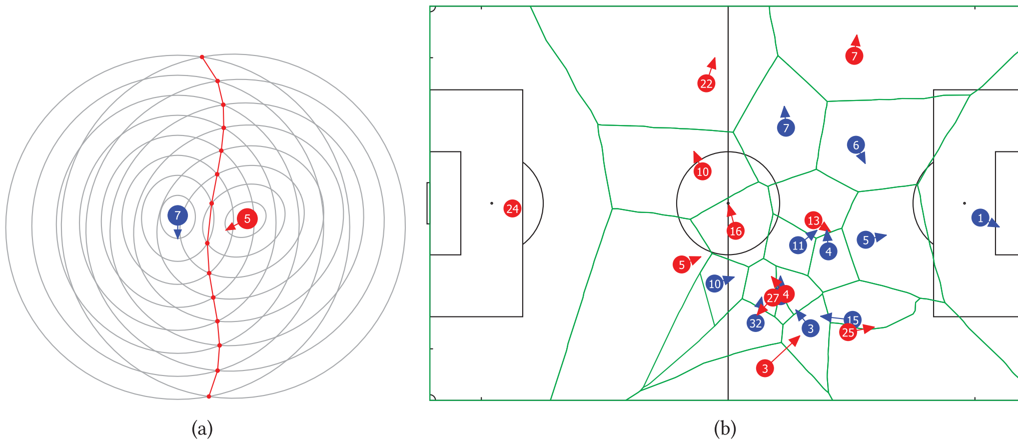


Fig. 8. (a) An approximate bisector between two players using the intersection points of the RPRs. (b) An example of the approximate dominant region subdivision by Gudmundsson and Wolle [2014].

time t. An advantage with using the RPR for computing dominant regions is that more complex motion models can be used by simply drawing the RPR for different values of t. They presented the following high-level algorithm. Given a sequence of timesteps t_i, $1 \leq i \leq k$, compute the RPRs for each player and each timestep. The algorithm then iterates through the sequence of timesteps, and, for each pair of players, the *partial dominant regions* are constructed from the RPRs. The partial dominant regions are then combined with the dominant regions computed in the previous timestep to form new dominant regions. Assuming that the RPR is a convex area for any p and any t, Nakanishi et al. claim a factor of 1,000 improvement in computation time at the cost of roughly a 10% drop in accuracy.

Gudmundsson and Wolle [2014] used RPRs induced from real trajectory data. They also presented an algorithm for constructing an approximate dominant region subdivision, which is superficially similar to the algorithm by Nakanishi et al. [2009]. However, instead of computing partial dominant regions for each pair of players at each timestep, an approximate bisector is constructed for every pair of players. An example of an approximate bisector between two players is shown in Figure 8(a), and in Figure 8(b) the final subdivision generated by the algorithm in Gudmundsson and Wolle [2014] is depicted.

A closer study of a player's dominant region was performed by Fonseca et al. [2012] in an attempt to describe the spatial interaction between players. They considered two variables denoting the smallest distance between two teammates and the size of the dominant region. They observed that the individual dominant regions seem to be larger for the attacking team. They also found that for the defending team the two measures were more irregular, which indicates that their movement was more unpredictable than the movement of the attacking team.

According to the authors, the player- and team-dominant regions detect certain match events such as "when the ball is received by an attacker inside the defensive structure, revealing behavioural patterns that may be used to explain the performance outcome."

Ueda et al. [2014] compared the team area and the -dominant region (within the team area) during offensive and defensive phases. The *team area* is defined as the smallest enclosing orthogonal box containing all the field players of the defending team. The results seem to show that there exists a correlation between the ratio of the dominant region to team area and the performance of the team's offence and defence. Dominant regions of successful attacks were thinner than those for unsuccessful attacks that broke down with a turnover event located near the centre of the playing area. The conclusion was that the dominant region is closely connected to the offensive performance, and, hence, perhaps it is possible to evaluate the performance of a group of players using the dominant region.

OPEN QUESTION 1. *The function modelling player motion used in dominant region computations has often been simple for reasons of tractability or convenience. Factors such as the physiological constraints of the players and a priori momentum have been ignored. A motion function that faithfully models player movement and is tractable for computation is an open problem.*

3.3.3. Further Applications. The dominant region is a fundamental structure that has been shown to support several other interesting measures andis discussed next.

(1) **(Weighted) Area of team-dominant region.** Taki et al. [1996] defined a *team-dominant region* as the union of dominant regions of all the players in the team. Variations in the size of the team-dominant region was initially regarded by Taki et al. [1996] as a strong indication on the performance of the team. However, Fujimura and Sugihara [2005] argued that the size of a dominant region does not capture the contribution of a player. Instead, they proposed using a *weighted* dominant region by weighting either with respect to the distance to the goal or with respect to the distance to the ball. They argued that both of these approaches better model the contribution of a player compared to simply using the size of the dominant region. However, no further analysis was performed.

    Fujimura and Sugihara [2005] also suggested that the weighted area of dominant regions can be used to evaluate attacking teamwork: tracking the weighted dominant region ("defensive power") over time for the defender marking each attacker will indicate the attacker's contribution to the team.

(2) **Passing evaluation.** A player's *passable area* is the region of the playing area where the player can potentially receive a pass. The size and the shape of the passable area depends on the motion model and the positions of the ball and the other players. Clearly, this is also closely related to the notion of dominant region.

    Definition 3.2 (*[Gudmundsson and Wolle 2014]*). A player p is open for a pass if there is some direction toward and (reasonable) speed with which the ball can be passed, such that p can intercept the ball before all other players.

    Taki and Hasegawa [2000] further classified a pass as "successful" if the first player who can receive the pass is a player from the same team. This model was extended and implemented by Fujimura and Sugihara [2005] as follows. They empirically developed a motion model for the ball, following Equation (1) in Section 3.3.1. They then defined the *receivable pass variation* for each player to be the number of passes the player can receive among a set of sampled passes. They experimentally sampled 54,000 passes by discretizing $[0, 2\pi)$ into 360 unit directions and speeds between 1 and 150km/h into 150 units.

    Gudmundsson and Wolle [2014] also used a discretization approach but viewed the problem slightly differently. Given the positions, speeds, and directions of motion of the players, they approximated who is open for a pass for each discretized ball speed. For each fixed passing speed, they built RPRs for each player and the ball over a set of discrete timesteps. Then an approximate bisector is computed between the ball and the player. Combining the approximate bisectors for all the players, a piecewise linear function f is generated over the domain $[0, 2\pi)$. The

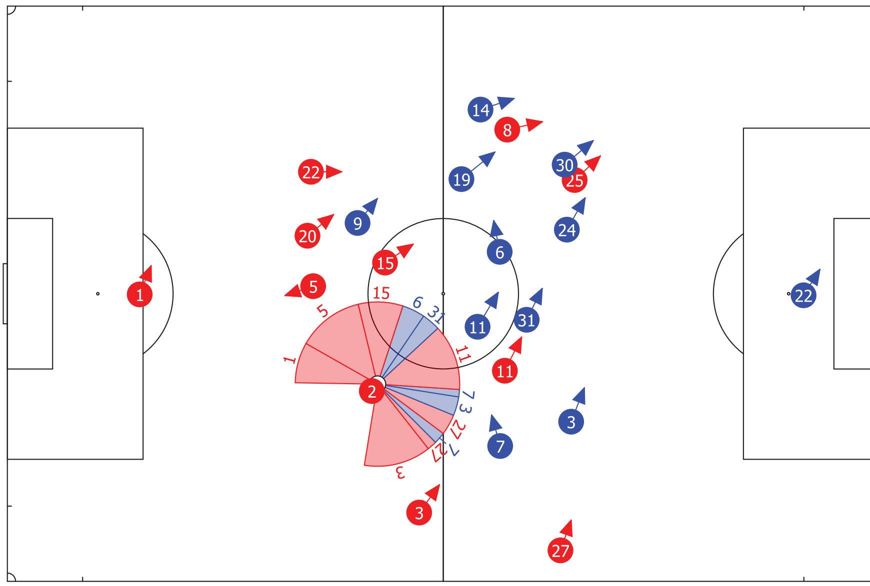


Fig. 9. Available receivers of a pass by player Red 2 where velocity of the ball is 20m/s. Each sector represents an interval on $[0, 2\pi)$ that indicates which player may receive the pass. Players may receive a pass made at more than one interval, for example, Blue 7.

segments of the bisectors that lie on the lower envelope of f map to intervals on the domain where the player associated with the bisector is open for a pass. An example of the output is shown in Figure 9 for a fixed ball speed.

Open Question 2. *The existing models for determining whether a player is open to receive a pass only consider passes made along the shortest path between passer and receiver and where the ball is moving at a constant velocity. The development of a more realistic model that allows for aerial passes, effects of ball spin, and variable velocities is an interesting research question.*

(3) **Spatial Pressure.** An important tactical measure is the amount of spatial pressure the team exerts on the opposition. Typically, when a team believes that the opponent is weak at retaining possession of the ball, a high pressure tactic is used. Taki et al. [1996] defined spatial pressure for a player p as follows:

$$m \cdot (1 - P) + (1 - m) \cdot (1 - d/D),$$

where, for a fixed radius r, P denotes the fraction of the disk of radius r with center at p that lies within the dominant region of opposing players, d is the distance between p and the ball, D is the maximal distance between p and any point on the playing area, and m is a preset weight. This definition was also used by Horton et al. [2015]. See Figure 10 for two examples of spatial pressure.

Open Question 3. *The definition of spatial pressure in Taki et al. [1996] is simple and does not model effects such as the direction the player is facing or the direction of pressuring opponents, both of which would intuitively be factors that ought to be considered. Can a model that incorporates these factors be devised and experimentally tested?*

(4) **Rebounding.** Traditionally a player's rebounding performance has been measured as the average number of rebounds per game. Maheswaran et al. [2014] presented a

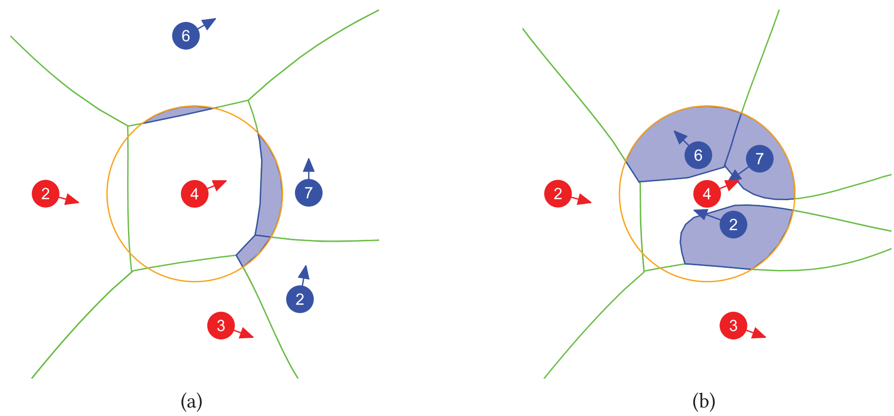


Fig. 10. Comparing the pressure that the encircled player is under in the two pictures shows that the encircled player on the right is under much more pressure.

model to quantify the potential to rebound unsuccessful shots in basketball in more detail. Simplified, the model considers three phases. The first phase is the *position* of the players when the shot is taken. From the time that the ball is released until it hits the rim, the players will try to move into a better position—the *crash* phase. After the crash phase, the players have the chance to make the rebound. The proficiency of a player in rebounding is the measured by the *conversion*—the third phase.

Both the positioning phase and the crash phase make use of the dominant region (Voronoi diagram) to value the position of the player; that is, they compute a "real estate" value of the dominant region of each player both when the shot is made and when the shot hits the rim. These values, together with the conversion, are combined into a *rebounding* value.

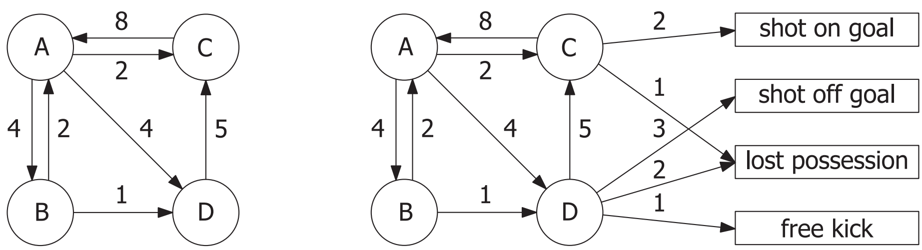
4. NETWORK TECHNIQUES FOR TEAM PERFORMANCE ANALYSIS

Understanding the interaction between players is one of the more important and complex problems in sports science. Player interaction can give insight into a team's playing style or be used to assess the importance of individual players to the team. Capturing the interactions between individuals is a central goal of social network analysis [Wasserman and Faust 1997], and techniques developed in this discipline have been applied to the problem of modelling player interactions.

An early attempt to use networks for sports analysis was in a study by Gould and Gatrell [1979], where they explore all passes made in the 1977 FA Cup Final between Liverpool and Manchester United. They studied the simplicial complexes of the passing network and made several interesting observations, including that the Liverpool team had two "quite disconnected" subsystems and that Kevin Keegan was "the linchpin of Liverpool." However, their analysis, while innovative, did not attract much attention.

Since the mid-2000s numerous articles have appeared that apply social network analysis to team sports. Two types of networks have dominated the research literature to date: *passing networks* and *transition networks*.

Passing networks have been the most frequently studied type in the research field. To the best of our knowledge, they were first introduced by Passos et al. [2011]. A passing network is a graph $G = (V, E)$, where each player is modelled as a vertex and two vertices v_1 and v_2 in V have a directed edge $e = (v_1, v_2)$ from v_1 to v_2 with integer weight $w(e)$ such that the player represented by vertex v_1 has made $w(e)$ successful



(a) (b)

Fig. 11. (a) A passing network modelling four players $\{A, B, C, D\}$ and the passes between the players. (b) A transition network is a passing network extended with outcomes. For example, twice player C made a shot on goal and once the player lost possession.

passes to the player represented by vertex v_2. A small example of a passing graph is shown in Figure 11(a). Passing networks can be constructed directly from *event logs*, defined in Section 2. A temporal sequence of passes made in a match is encoded as a path within the passing network. A passing network that is extended with outcomes, as illustrated in Figure 11(b), is then referred to as a *transition network*.

Many properties of passing networks have been studied, among them *density*, *heterogeneity*, *entropy*, and *Nash equilibria*. However, the most-studied measurement is *centrality*. We begin by considering centrality and its variants, and then we briefly consider some of the other measures discussed in the literature.

4.1. Centrality

Centrality measures were introduced in an attempt to determine the key nodes in a network, for example, to identify the most popular persons in a social network or super-spreaders of a disease [Newman 2010]. In team sports, the aim of using centrality measurements is generally to identify key players or to estimate the interactivity between team members. For an excellent survey on network centrality; see Borgatti [2005].

4.1.1. Degree Centrality. The simplest centrality measure is the *degree centrality*, which is the number of edges incident to a vertex. For directed networks, one usually distinguishes between the in-degree and the out-degree centrality. In sports analysis, the out-degree centrality is simply referred to as *centrality*, while the in-degree centrality is usually called the *prestige* of a player. Some articles consider both centrality and prestige, see, for example, Clemente et al. [2015b], but most of the literature has focused on centrality.

Fewell et al. [2012] considered a transition graph induced from basketball games where the vertices represented the five traditional player positions (point guard, shooting guard, small forward, power forward, and center), possession origins, and possession outcomes. The centrality was computed on the transition graph and split into two outcomes: “shots” and “others.” The measure was computed on 32 basketball games, and prior knowledge about the importance of players to the teams involved was compared to the centrality values of the players. They used degree centrality to compare teams that heavily rely on key players with teams with a more even distribution between their team members. Unfortunately, the data were not definitive, since the overall centrality rankings did not show a strong relation to the teams performance.

Grund [2012] used degree centrality together with Freeman centralization [Freeman 1978]. The idea by Freeman was to consider the relative centrality of the most important node in the network. That is, how central is the most central node compared to the centrality of the other nodes in the network. The Freeman centrality is measured as the sum of the differences between the node with the highest degree centrality and all other nodes and then divided by a value depending only on the size of the network [Freeman 1978]. They used an extensive set of 283,259 passes from 760 English Premier League football matches for their experiments. From a team performance perspective, Grund [2012] set out to answer two hypotheses: (i) increased interaction between players leads to increased team performance, and (ii) increased interaction centralization leads to decreased team performance. The latter is strongly connected to centrality, and Grund [2012] went on to show that a high level of centralization decreases team performance.

In a series of recent articles, Clemente et al. [2014, 2015a, 2015b, 2015] argue that centrality may recognise how football players collaborate and the nature and strength of their collaboration. For example, central midfielders and central defenders usually show higher degree centrality then other players. Some exceptions were shown in Clemente et al. [2014b], where the left and right defenders also obtained very high degree centrality. In general, goal-keepers and forwards have the lowest centrality measure.

4.1.2. Betweenness Centrality. The betweeness centrality of a node is the number of times it lies on the shortest path between two other nodes in the network. Originally, it was introduced by Freeman [1977] in an attempt to estimate "a human's potential control of communication in a social network."

Peña and Touchette [2012] claimed that the betweenness centrality measures how the ball flow between football players depends on a particular player and, as such, provides a measure of the impact of the "removal" of that player from the game, either by being sent off or by being isolated by the opponents. They also argued that, from a tactical point of view, a team should aim to have a balanced betweenness score for all players.

A centrality measure closely related to the betweenness centrality is flow centrality. The flow centrality is measured by the proportion of the entire flow between two vertices that occur on paths of which a given vertex is a part.

Duch et al. [2010] considered flow centrality for transition networks where the weight of an edge from a player v_1 to a player v_2 is equal to the fraction of passes initiated by v_1 to reach v_2. Similarly, the shooting accuracy for a player (the weight of the edge from the player to the vertex "shots on goal") is the fraction of shots made by the player that end up on goal. They then studied the flow centrality over all paths that results in a shot in football. They also take the defensive performance into account by having each player initiate a number of flow paths, which is comparable to the number of times the player wins possession of the ball. The *match performance* of the player is then the normalised value of the logarithm of this combined value. They argue that this gives an estimate of the contribution of a single player and also of the whole team. The team's match performance value is the mean of the individual player values. Using these values, both for teams and for individual players, Duch et al. [2010] analysed 20 games from the football 2008 Union Européenne de Football Association (UEFA) European Cup. They claim that their measurements provide "sensible results that are in agreement with the subjective views of analysts and spectators"; in other words, the better-paid players tend to contribute more to the team's performance.

4.1.3. Closeness Centrality. The standard distance metric used in a network is the length (weight or cost) of the shortest path between pairs of nodes. The *closeness centrality* of a node is defined as the inverse of the *farness* of the node, which is the sum of its distance to all other nodes in the network [Bavelas 1950]. Peña and Touchette [2012] argued that the closeness score is an estimate of how easy it is to get the ball to a specific player; that is, a high closeness score indicates a well-connected player within the team. They made a detailed study using passing data from the 2010 Fédération

Internationale de Football Association (FIFA) World Cup. The overall conclusion they reached was that there is a high correlation between high scores in closeness centrality, *PageRank*, and clustering (see below), which supports the general perception of the players performance reported in the media at the time of the tournament.

4.1.4. Eigenvector Centrality and PageRank. The general idea of Eigenvector centrality and PageRank is that the importance of a node depends not only on its degree but also on the importance of its neighbours. Cotta et al. [2013] used the eigenvector centrality calculated with the power iteration model by von Mises and Pollaczek-Geiringer [1929]. The measure aims to identify which player has the highest probability to be in possession of the ball after a sequence of passes. They also motivated their measure by a thorough analysis of three games from the 2010 FIFA World Cup, where they argued the correlation between the eigenvector centrality score and the team's performance.

A variant of the eigenvector centrality measure is *PageRank*, which was one of the algorithms used by Google Search to rank web pages [Brin and Page 1998]. The passing graph is represented as an adjacency matrix A where each entry A_{ji} is the number of passes from player j to player i. In football terms, the *PageRank* centrality index for player i is defined as

$$x_i = p \sum_{j \neq i} \frac{A_{ji}}{L_j^{out}} \cdot x_j + q,$$

where $L_j^{out} = \sum_k A_{jk}$ is the total number of passes made by player j, p is the parameter representing the probability that a player will decide to give the ball away rather than keep it and shoot, and q is a "free" popularity assigned to each player. Note that the *PageRank* score of a player is dependant on the scores of the player's team mates. Peña and Touchette [2012] argue that the *PageRank* measure gives each player a value that is approximately the likelihood that the player will be in possession of the ball after a fixed number of passes. Using data from the 2010 FIFA World Cup, they computed the *PageRank* for the players in the top 16 teams but focused their discussion on the players in the top four teams: Spain, Germany, Uruguay, and the Netherlands. They showed that the *PageRank* of players in the Dutch and Uruguay teams were more evenly distributed than players from Spain and Germany. This indicates that no player on those teams has a predominant role in the passing scheme, while Xavi Hernandez (Spain) and Bastian Schweinsteiger (Germany) were particularly central to their teams.

4.2. Clustering Coefficients

A clustering coefficient is a measure of the degree of which nodes in a network are inclined to cluster together. In the sport science literature, both the *global* and the *local* clustering coefficients have been applied. The idea of studying the global cluster coefficient of the players in a team is that it reflects the cooperation between players, that is, the higher the coefficient for a player, the higher is his or her cooperation with the other members of the team [Clemente et al. 2014; Fewell et al. 2012; Peña and Touchette 2012]. Fewell et al. [2012] also argued that a high global clustering coefficient indicates that attacking decisions are taken by several players, which increases the number of possible attacking paths that have to be assessed by defences in basketball. Peña and Touchette [2012] showed, using the 2010 FIFA World Cup passing data, that Spain, Germany, and the Netherlands consistently had very high clustering scores when compared to Uruguay, suggesting that they were extremely well-connected teams, in the sense that almost all players contribute.

Cotta et al. [2013] considered three games involving Spain from the 2010 FIFA World Cup and used the local clustering coefficient as a player coefficient. They studied how the coefficient changed during the games, and argued for a correlation between the number of passes made by Spain and the local clustering coefficient. They claimed that Spain's clustering coefficient remains high over time, "indicating the elaborate style of the Spanish team."

It should be noted that it is not completely clear that there is a strong connection between the clustering coefficient and the team performance. For example, Peña and Touchette [2012] stated that in their study they did not get any reasonable results and "will postpone the study of this problem for future work."

OPEN QUESTION 4. *Various centrality and clustering measures have been proposed to accurately represent some aspect of player or team performance. A systematic study reviewing all such measures against predefined criteria, and on a large dataset, would be a useful contribution to the field.*

4.3. Density and Heterogeneity

In general, it is believed that stronger collaboration between football players (i.e., more passes) will make the team stronger—known as the *density-performance hypothesis* [Balkundi and Harrison 2006]. Therefore, a widely assessed measure of networks is density, which is traditionally calculated as the number of edges divided by the total number of possible edges. This is the density measure used by Clemente et al. in a series of recent articles [2014a, 2015a, 2015b, 2015]. For weighted graphs, the measurement becomes slightly more complex. Grund [2012] defined the *intensity* of a team as the sum of the weighted degrees over all players divided by the total time the team have possession of the ball, that is, possession-weighted passes per minute.

Related to the density is *passing heterogeneity*, which Cintia et al. [2015] defined as the standard deviation of the vertex degree for each player in the network. High heterogeneity of a passing network means that the team tends to coalesce into subcommunities and that there is a low level of cooperation between players [Clemente et al. 2015]. One interesting observation made by Clemente et al. [2015] was that the density usually went down in the second half while the heterogeneity went up.

OPEN QUESTION 5. *The density-performance hypothesis suggests an interesting metric of team performance. Can this hypothesis be tested scientifically?*

4.4. Entropy, Topological Depth, Price-of-Anarchy, and Power-Law Distributions

As described above, Fewell et al. [2012] considered an extended transition graph for basketball games, where they also calculated *player entropy*. Shannon entropy [Shannon 2001] was used to estimate the uncertainty of a ball transition. The *team entropy* is the aggregated player entropies, which can be computed in many different ways. Fewell et al. [2012] argue that, from the perspective of the opposing team, the real uncertainty is the number of options and computed the team entropy from the transition matrix describing ball movement probabilities across the five standard player positions and the two outcomes.

Skinner [2010] showed that passing networks have two interesting properties. They identified a correspondence between a basketball transition network and a traffic network and used insights from the latter to make suppositions about the former. They posited that there may be a difference between the Nash equilibrium of a transition network and the community optimum—the *Price of Anarchy*. In other words, for the best outcome, one should not always select the highest-percentage shot. A similar observation was made in Fewell et al. [2012], who noted that the low flow centrality of the most utilised position (point guard) seems to indicate that the contribution of key players can be negatively affected by the point guard controlling the ball more often than other players. Related to the same concept, Skinner [2010] suggested that removing a key player from a match—and hence the transition network—may actually *improve* the team performance, a phenomena known as the *Braess' paradox* in network analysis [Braess et al. 2005].

5. DATA MINING

The representations and structures described in Sections 2–4 are informative in isolation but may also be the input for more complex algorithmic and probabilistic analysis of team sports. In this section, we present a task-oriented survey of the techniques that have been applied and outline the motivations for these tasks.

5.1. Applying Labels to Events

Sports analysts are able to make judgments about events and situations that occur in a match and apply qualitative or quantitative attributes to that event, for example, to rate the riskiness of an attempted shot on goal or the quality of a pass. Event labels such as these can be used to measure player and team performance and are currently obtained manually by video analysis. Algorithmic approaches to automatically produce such labels may improve the efficiency of the process.

Horton et al. [2015] presented a classifier that determines the quality of passes made in football matches by applying a label of *good*, *OK*, or *bad* to each pass made and were able to obtain an accuracy rate of 85.8%. The classifier uses features that are derived from the spatial state of the match when the pass occurs, including features derived from the dominant region described in Section 3.3, which were found to be important features to the classifier.

In research by Beetz et al. [2009], the approach was to cluster passes and to then induce a decision tree on each cluster where the passes were labelled as belonging to the cluster or not. The feature predicates, learned as splitting rules, in the tree could then be combined to provide a description of the important attributes of a given pass.

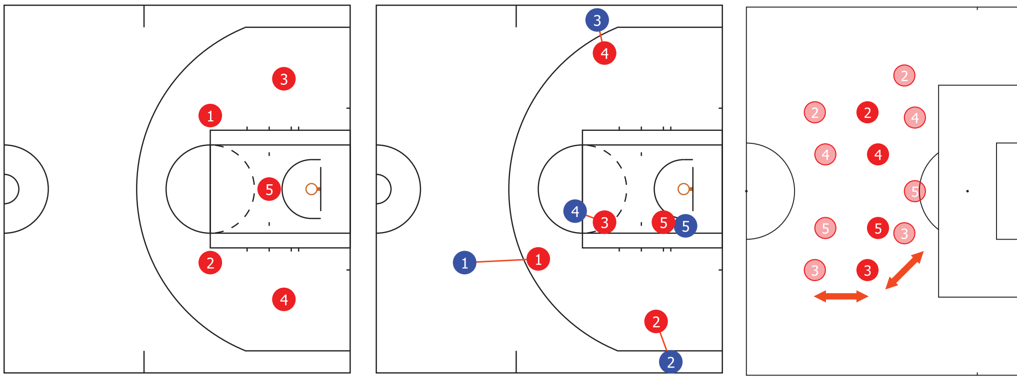
Bialkowski et al. [2014b] used the formation descriptors computed with the algorithm presented in Bialkowski et al. [2014c] (see Section 5.3) to examine whether formations could accurately predict the identity of a team. In the model, a linear discriminant analysis classifier was trained on features describing the team formation, and the learned model was able to obtain an accuracy of 67.23% when predicting a team from a league of 20 teams.

In Maheswaran et al. [2012], the authors perform an analysis of various aspects of the rebound in basketball to produce a rebound model. The rebound is decomposed into three components: the location of the shot attempt, the location where the rebound is taken, and the height of the ball when the rebound is taken. Using features derived from this model, a binary classifier was trained to predict whether a missed shot would be successfully rebounded by the offensive team. The model was evaluated and obtained an accuracy rate of 75% in experiments on held-out test data.

5.2. Predicting Future Event Types and Locations

The ability to predict how play will unfold given the current game state has been researched extensively, particularly in the computer vision community. This has an application in automated camera control, where the camera filming a match must automatically control its pitch, tilt, and zoom. The framing of the scene should ideally contain not just the current action but also the movement of players who can be expected to be involved in future action and the location of where such future action is likely to occur.

Kim et al. [2010] considered the problem of modelling the evolution of football play from the trajectories of the players, such that the location of the ball at a point in the



(a) Zone Defence (b) Man-marking Defence (c) Back-four Defence

Fig. 12. Examples of typical formations used in basketball and football. (a) The zone defence is spatially anchored to the dimensions of the court, and the players' positioning is invariant to the phase of play. (b) Defenders who are man-marking will align themselves relative to their opposing player, typically between the attacker and the basket. (c) The back-four formation in football maintains the alignment of players in the formation but will move forward and laterally, depending on the phase of play.

near future could be predicted. Player trajectories were used to compute a dense motion field over the entire playing area, and points of convergence within the motion field were identified. The authors suggest that these points of convergence indicate areas where the ball can be expected to move to with high probability, and the experiments described in the article demonstrate this with several examples.

Yue et al. [2014] construct a model to predict whether a basketball player will shoot, pass to one of four teammates, or retain possession. The action a player takes is modelled using a multi-class conditional random field. The input features to the classifier include latent factors representing player locations that are computed using non-negative matrix factorization—see Section 3.1—and the experimental results show that these features improve the predictive performance of the classifier.

Wei et al. [2015] constructed a model to make short-term predictions of which football player will be in possession of the ball after a given interval. They propose a model—augmented-Hidden Conditional Random Fields—that combines local observation features with the hidden layer states to make the final prediction of the player who possess the ball. The experimental results show that they are able to design a model that can predict which player will be in possession of the ball after 2s with 99.25% accuracy.

5.3. Identifying Formations

Sports teams use pre-devised spatial formations as a tactic to achieve a particular objective. The ability to automatically detect such formations is of interest to sports analysts and coaches. For example, a coach would be interested in understanding the proportion of time that a team maintains an agreed-on formation and when the team is compelled by the circumstances of the match to alter its formation. Moreover, when preparing for a future opponent, an understanding of the formation used and periods where the formation changes would be of interest.

A formation is a positioning of players, relative to the location of other objects, such as the pitch boundaries or goal/basket, the players' teammates, or the opposition players. Formations may be spatially anchored, as in, for example, a zone defence in basketball where players position themselves in a particular location on the playing area, see Figure 12(a). On the other hand, a formation may vary spatially but maintain a stable relative orientation between the players in the formation. For example, the defensive players in a football team will position themselves in a straight line across the pitch, and this line will move as a group around the pitch, depending on the phase of play, Figure 12(c). Finally, a different type of formation is a *man marking* defence, where defending players will align themselves relative to the attacking players that they are marking, Figure 12(b). In this scenario, the locations of defenders may vary considerably, relative to their teammates or to the boundaries of the playing area.

Moreover, the players that fulfil particular roles within a formation may switch, either explicitly, through substitutions, or dynamically where players may swap roles for tactical reasons. The following approaches have been used to determine formations from the low-level trajectory signal.

Lucey et al. [2013] investigated the assignment of players to roles in field hockey, where teams use a formation of three lines of players arrayed across the field. At any time t there is a one-to-one assignment of players to roles; however, this assignment may vary from timestep to timestep. This problem is mathematically equivalent to permuting the player ordering $\mathbf{p}_t^\tau$ using a permutation matrix $\mathbf{x}_t^\tau$ that assigns the players to roles $r_t^\tau = \mathbf{x}_t^\tau \mathbf{p}_t^\tau$. The optimal permutation matrix $\mathbf{x}_t^\tau$ should minimise the total Euclidean distance between the reference location of each role and the location of the player assigned to the role and can be computed in closed form using the Hungarian algorithm [Kuhn 1955].

Wei et al. [2013] used this approach as a preprocessing step on trajectory data from football matches, and the computed role locations were subsequently used to temporally segment the matches into game phases. Lucey et al. [2014a] applied role assignment to basketball players in sequences leading up to three-point shots. They analysed close to 20,000 such shots and found that role-swaps involving particular pairs of players in the moments preceding a three-point shot have a significant impact on the probability of the shooter being *open*—at least 6 feet away from the nearest marker—when the shot is made.

Furthermore, Bialkowski et al. [2014c] observed that the role assignment algorithm presented by Lucey et al. [2013] required a predefined prototype formation to which the players are assigned. They consider the problem of simultaneously detecting the reference location of each role in the formation, and assigning players to the formation, using an expectation maximization approach [Dempster et al. 1977]. The initial role reference locations are determined as the mean position of each player. The algorithm then uses the Hungarian algorithm to update the role assignment for each player at each timestep, and then the role reference locations are recomputed according to the role assignment. The new locations are used as input for the next iteration, and the process is repeated until convergence.

The learned formations for each team and match were then clustered into six formations, and the authors claim that the clustered formations were consistent with expert knowledge of formations used by football teams. This was validated experimentally by comparing the computed formation with a formation label assigned by an expert, and an accuracy of 75.33% was obtained.

In a subsequent article, Bialkowski et al. [2014a] investigated differences in team strategies when playing home or away by using formations learned with the role assignment algorithm. By computing the mean position when teams are playing at home from when they are playing away, they observed that teams defend more deeply when away from home; in other words, they set their formation closer to the goal they are defending.

A qualitatively different formation is for players to align themselves with the positions of the opposition players, such as the *man-marking* defense used in basketball; see Figure 12(b). Franks et al. [2015a] defined a model to determine which defender is marking each attacker. For a given offensive player at a given time, the mean location of the defender is modelled as a convex combination of three locations: the position of the attacker, the location of the ball, and the location of the hoop. The location of a defender, given the observed location of the marked attacker, is modelled as a Gaussian distribution about a mean location. The matching between defenders and the attacker that they are marking over a sequence of timesteps is modelled using a Hidden Markov Model, ensuring that the marking assignments are temporally smoothed.

5.4. Identifying Plays and Tactical Group Movement

Predefined *plays* are used in many team sports to achieve some specific objective. American football uses highly structured plays where the entire team has a role and their movement is highly choreographed. On the other hand, plays may also be employed in less structured sports, such as football and basketball, when the opportunity arises, such as the *pick and roll* in basketball or the *one-two* or *wall pass* in football. Furthermore, teammates who are familiar with each other's playing style may develop ad hoc productive interactions that are used repeatedly, a simple example of which is a sequence of passes between a small group of players. Identification of plays is a time-consuming task that is typically carried out by a video analyst, and thus a system to perform the task automatically would be useful.

An early attempt in this direction attempted to recognise predefined plays in American football [Intille and Bobick 1999]. They model a play as a choreographed sequence of movements by attacking players, each trying to achieve a local goal and in combination achieve a group goal for the play. The approach taken was to encode predefined tactical plays using a temporal structure description language that described a local goal in terms of a sequence of actions carried out by an individual player. These local goals were identified in the input trajectories using a Bayesian belief network. A second belief network then identified whether a global goal had been achieved based on the detected local goals—signifying that the play has occurred.

Two articles by Li et al. investigated the problem of identifying group motion, in particular the type of offensive plays in American football. Li et al. [2009] presented the Discriminative Temporal Interaction Network (DTIM) framework to characterise group motion patterns. The DTIM is a temporal interaction matrix that quantifies the interaction between objects at two given points in time. For each predefined group motion pattern—a play—a multi-modal density was learned using a properly defined Riemannian metric, and a *Maximum a Posteriori* (MAP) classifier was then used to identify the most likely play for a given input set of trajectories. The experiments demonstrated that the model was able to accurately classify sets of trajectories into five predefined plays and outperformed several other common classifiers for the task. This model has the advantage of not requiring an *a priori* definition of each player's movement in the play, as required in Intille and Bobick [1999].

Li and Chellappa [2010] considered group motion segmentation, where a set of unlabelled input trajectories are segmented into the subset that participated in the group motion and those that did not. The problem was motivated by the example of segmenting a set of trajectories into the set belonging to the offensive team (who participated in the play) and the defensive team (who did not). The group motion is modelled as a dynamic process driven by a *spatio-temporal driving force*—a densely distributed motion field over the playing area. The driving force is modelled as a 3×3 matrix $F(t_0, t_f, x, y)$ such that $X(t_f) = F(t_0, t_f, x, y)X(t_0)$. Thus, an object located at $X(t_0)$ at time t_0 will be driven to $X(t_f)$ at time t_f. Using Lie group theory [Rossmann 2002], a Lie algebraic representation f of F is determined with the property that the space of all fs is linear, and thus tractable statistical models can be induced from f. A Gaussian mixture model was used to learn a fixed number of driving forces at each timestep, which was then used to segment the trajectories.

There has been a number of diverse efforts to identify commonly occurring sequences of passes in football matches. In Borrie et al. [2002], the pitch is subdivided into zones, and sequences of passes are identified by the zones that they start and terminate in. A possession can thus be represented by a string of codes representing each pass by source and target zone, and with an elapsed time between them. They introduce *T-pattern* analysis, which is used to compute possessions where the same sequence of passes are made with consistent time intervals between each pass, and frequently occurring patterns could thus be identified. Camerino et al. [2012] also used T-pattern analysis on pass strings; however, the location of passes was computed relative to the formation of the team in possession, for example, between the defense and midfield, or in front of the attacking line.

An algorithm to detect frequently occurring sequences of passes was presented in Gudmundsson and Wolle [2014]. A suffix tree [Weiner 1973] was used as a data structure D to store sequences of passes between individual players. A query (τ, o) can then be made against D that returns all permutations of τ players such that the ball is passed from a player p_1 to p_τ, via players $p_2, \ldots, p_{\tau-1}$ at least o times, and thus determines commonly used passing combinations between players.

Van Haaren et al. [2015] considered the problem of finding patterns in offensive football strategies. The approach taken was to use inductive logic programming to learn a series of clauses describing the pass interactions between players during a possession sequence that concludes with a shot on goal. The passes were characterised by their location on the pitch, and a hierarchical model was defined to aggregate zones of the pitch into larger regions. The result is a set of rules, expressed in first-order predicate logic, describing the frequently occurring interaction sequences.

Research by Wang et al. [2015] also aimed to detect frequent sequences of passing. They claim that the task of identifying tactics from pass sequences is analogous to identifying topics from a document corpus and present the Team Tactic Topic Model (T^3M) based on Latent Dirichlet Allocation [Blei et al. 2003]. Passes are represented as a tuple containing an order-pair of the passer and receiver and a pair of coordinates representing the location where the pass was received. The T^3M is an unsupervised approach for learning common tactics, and the learned tactics are coherent with respect to the location where they occur and the players involved.

5.5. Temporally Segmenting the Game

Segmenting a match into phases based on a particular set of criteria is a common task in sports analysis, as it facilitates the retrieval of important phases for further analysis. The following paragraphs describe approaches that have been applied this problem for various types of criteria.

Hervieu et al. [2009, 2011] present a framework for labelling phases within a handball match from a set of predefined labels representing common attacking and defensive tactics. The model is based on a hierarchical parallel semi-Markov model and is intended to model the temporal causalities implicit in player trajectories. In other words, modelling the fact that one player's movement may cause another player to subsequently alter his or her movement. The upper level of the hierarchical model is a semi-Markov model with a state for each of the defined phase labels, and within each state the lower level is a parallel hidden Markov model for each trajectory. The duration of time spent in each upper level state is modelled using a Gaussian mixture model. In the experiments, the model was applied to a small dataset of handball match trajectories from the 2006 Olympics Games final and resulted in accuracy of 92% on each timestep compared to the ground truth provided by an expert analyst. The model exactly predicted the sequence of states, and the misclassifications were all the result of time lags when transitioning from one state to the subsequent state.

Perše et al. [2009] investigated segmentation of basketball matches. A framework with two components was used; the first segmented the match duration into sequences of offensive, defensive, or time-out phases. The second component identified basic activities in the sequence by matching to a library of predefined activities, and the sequences of activities were then matched with predefined templates that encoded known basketball plays.

Wei et al. [2013] considered the problem of automatically segmenting football matches into distinct game phases that were classified according to a two-level hierarchy using a decision forest classifier. At the top level, phases were classified as being *in-play* or a *stoppage*. *In-play* phases were separated into highlights or non-highlights, and *stoppages* were classified by the reason for the stoppage: *out for corner*, *out for throw-in*, *foul*, or *goal*. The classified sequences were subsequently clustered to find a team's most probable method of scoring and of conceding goals.

In a pair of articles by Bourbousson et al. [2010a, 2010b], the spatial dynamics in basketball was examined using relative-phase analysis. In Bourbousson et al. [2010a], the spatial relation between dyads of an attacking player and their marker were analysed. In Bourbousson et al. [2010b], the pairwise relation between the centroid of each team was used, along with a *stretch index* that measured the aggregate distance betweens players and their team's centroid. A Hilbert transformation was used to compute the relative phase in the x and y directions of the pairs of metrics. Experimental results showed a strong in-phase relation between the various pairs of metrics in the matches analysed, suggesting individual players and also teams move synchronously. The authors suggest that the spatial relations between the pairs are consistent with their prior knowledge of basketball tactics.

Frencken et al. [2011] performed a similar analysis of four-a-side football matches. They used the centroid and the convex hull induced by the positions of the players in a team to compute metrics, for example, the distance in the x and y directions of the centroid, and the surface area of the convex hull. The synchronized measurements for the two teams were modelled as coupled oscillators, using the *Haken-Kelso-Bunz* (HKB) model [Haken et al. 1985]. Their hypothesis was that the measurements would exhibit in-phase and anti-phase coupling sequences and that the anti-phase sequences would denote game-phases of interest. In particular, the authors claim that there is a strong linear relationship between the x direction of the centroid of the two teams and that phases where the centroid's x directions cross are indicative of unstable situations that are conducive to scoring opportunities. They note that such a crossing occurs in the buildup to goals in about half the examples.

OPEN QUESTION 6. *Coaches and analysts are often interested in how the intensity of a match varies over time, as periods of high intensity tend to present more opportunities and threats. It is an interesting open problem to determine if it is possible to compute a measure of intensity from spatio-temporal data and thus be able to determine high-intensity periods.*

6. PERFORMANCE METRICS

Determining the contribution of the offensive and defensive components of team play has been extensively researched, particularly in the case of basketball, which has several useful properties in this regard. For example, a basketball match can be easily segmented into a sequence of *possessions*—teams average around 92 possessions per game [Kubatko et al. 2007]—most of which end in a shot at goal, which may or may not be successful. This segmentation naturally supports a variety of offensive and defensive metrics [Kubatko et al. 2007]; however, the metrics are not spatially informed, and, intuitively, spatial factors are significant when quantifying both offensive and defensive performance. In this section, we survey a number of research articles that use spatio-temporal data from basketball matches to produce enhanced performance metrics.

6.1. Offensive Performance

Shooting effectiveness is the likelihood that a shot made will be successful, and *effective field goal percentage* (EFG) is a de-facto metric for offensive play in basketball [Kubatko et al. 2007]. However, as Chang et al. [2014] observe, this metric confounds the efficiency of the shooter with the difficulty of the shot. Intuitively, spatial factors, such as the location where a shot was attempted from, and the proximity of defenders to the shooter would have an impact of the difficulty of the shot. This insight has been the basis of several efforts to produce metrics that provide a more nuanced picture of a player or team's shooting efficiency.

Early work in this area by Reich et al. [2006] used shot chart data (a list of shots attempted, detailing the location, time, shooter, and outcome of each shot). The article contained an in-depth analysis of the shooting performance of a single player—Sam Cassell of the Minnesota Timberwolves—over the entire 2003/2004 season. A vector of Boolean-valued predictor variables was computed for each shot, and linear models were fitted for shot frequency, shot location, and shot efficiency. By fitting models on subsets of the predictor variables, the authors analysed the factors that were important in predicting shot frequency, location, and efficiency.

Miller et al. [2014] investigated shooting efficiency by using vectors computed with non-negative matrix factorization to represent spatially distinct shot types; see Section 3.2. The shooting factors were used to estimate spatial shooting efficiency surfaces for individual players. The efficiency surfaces could then be used to compute the probability of a player making a shot conditioned on the location of the shot attempt, resulting in a spatially varying shooting efficiency model for each individual player.

Cervone et al. [2014] present *expected possession value* (EPV), a continuous measure of the expected points that will result from the current possession. EPV is thus analogous to a "stock ticker" that provides a valuation of the possession at any point in time during the possession. The overall framework consists of a macro-transition model that deals with game-state events such as passes, shots, and turnovers, and a micro-transition model that describes player movement within a phase when a single player is in possession of the ball. Probability distributions, conditioned on the spatial layout of all players and the ball, are learned for the micro- and macro-transition models. The spatial effects are modelled using non-negative matrix factorization to provide a compact representation that the authors claim has the attributes of being computationally tractable with good predictive performance. The micro- and macro-transition models are combined in a Markov chain, and from this the expected value of the end-state—scoring 0, 2, or 3 points—can be determined at any time during the possession. Experimental results in the article show how the EPV metric can support a number of analyses, such as *EPV-Added*, which compares an individual player's offensive value with that of a league-average player receiving the ball in the same situation, or *Shot Satisfaction*, which quantifies the satisfaction (or regret) of a player's decision to shoot rather than taking an alternative option such as passing to a teammate.

Chang et al. [2014] introduces another spatially informed measure of shooting quality in basketball: *Effective Shot Quality* (ESQ). This metric measures the value of a shot taken by a league-average player. ESQ is computed using a learned least-squares regression function whose input includes spatial factors such as the location of the shot attempt and the proximity of defenders to the shooter. Furthermore, the authors introduce EFG+, which is calculated by subtracting ESQ from EFG. EFG+ is thus an estimate of how well a player shoots relative to expectation, given the spatial conditions under which the shot was taken.

A further metric, *Spatial Shooting Effectiveness*, was presented by Shortridge et al. [2014]. Using a subdivision of the court, an empirical Bayesian scoring rate estimator was fitted using the neighbourhood of regions to the shot location. The spatial shooting effectiveness was computed for each player in each region of the subdivision and is the difference between the points-per-shot achieved by the player in the region and the expected points-per-shot from the estimator. In other words, it is the difference between a player's expected and actual shooting efficiency and thus measures how effective a player is at shooting relative to the league-average player.

Lucey et al. [2014b] considered shooting efficiency in football. They make a similar observation that the location where a shot is taken significantly impacts the likelihood of successfully scoring a goal. The proposed model uses logistic regression to estimate the probability of a shot succeeding—the *Expected Goal Value*. The input features are based on the proximity of defenders to the shooter and to the path the ball would take to reach the goal, the location of the shooter relative to the lines of players in the defending team's formation, and the location from which the shot was taken. The model is empirically analysed in several ways. The number of attempted and successful shots for an entire season is computed for each team in a professional league and compared to the expected number of goals that the model predicts, given the chances. The results are generally consistent, and the authors are able to explain away the main outliers. Furthermore, matches where the winning team has fewer shots at goal are considered individually, and the expected goals under the model are computed. This is shown to be a better predictor of the actual outcome, suggesting that the winning team was able to produce fewer—but better—quality chances.

6.2. Defensive Performance

Measures of defensive performance have traditionally been based on summary statistics of *interventions* such as blocks and rebounds in basketball [Kubatko et al. 2007] and tackles and clearances in football. However, Goldsberry and Weiss [2013] observed that, in basketball, the defensive effectiveness ought to consider factors such as the spatial dominance by the defence of areas with high rates of shooting success; the ability of the defence to prevent a shot from even being attempted; and secondary effects in the case of an unsuccessful shot, such as being able to win possession or being well-positioned to defend the subsequent phase.

In order to provide a finer-grained insight into defensive performance, Goldsberry and Weiss [2013] presented *spatial splits* that decompose shooting frequency and efficiency into a triple consisting of close-range, mid-range, and 3-point-range values. The offensive half-court was subdivided into three regions, and the shot frequency and efficiency were computed separately for shots originating in each region. These offensive metrics were then used to produce defensive metrics for the opposing team by comparing the relative changes in the splits for shots that an individual player was defending to the splits for the league-average defender.

An alternate approach to assessing the impact of defenders on shooting frequency and efficiency was taken by Franks et al. [2015a, 2015b]. They proposed a model that quantifies the effectiveness of man-to-man defense in different regions of the court. The proposed framework includes a model that determines who is marking whom by assigning each defender to an attacker. For each attacker, the canonical position for the defender is computed based on the relative spatial location of the attacker, the ball, and the basket. A hidden Markov model is used to compute the likelihood of an assignment of defenders to attackers over the course of a possession, trained using the expectation maximization algorithm [Dempster et al. 1977]. A second component of the model learned spatially coherent shooting type bases using non-negative matrix factorization on a shooting intensity surface fitted using a log-Gaussian Cox process. By combining the assignment of markers and the shot type bases, the authors were able to investigate the extent to which defenders inhibit (or encourage) shot attempts in different regions of the court and the degree to which the efficiency of the shooter is affected by the identity of the marker.

Another aspect of defensive performance concerns the actions when a shot is unsuccessful, and both the defence and attack will attempt to *rebound* the shot to gain possession. This was investigated by Maheswaran et al. [2014], where they deconstructed the rebound into three components: *positioning*, *hustle,* and *conversion*, described in Section 3.3.2. Linear regression was used to compute metrics for player's *hustle* and *conversion*, and experimental results showed that the top-ranked players on these metrics were consistent with expert consensus of top-performing players.

On the other hand, Wiens et al. [2013] performs a statistical evaluation of the options that players in the offensive team have when a shot is made in basketball. Players near the basket can either *crash the boards*—move closer to the basket in anticipation of making a rebound—or retreat in order to maximise the time to position themselves defensively for the opposition's subsequent attack. The model used as factors the players' distance to the basket and proximity of defenders to each attacking player. The experimental results suggested that teams tended to retreat more than they should, and thus a more aggressive strategy could improve a team's chances of success.

The analysis of defense in football would appear to be a qualitatively different proposition, in particular because scoring chances are much less frequent. To our knowledge, similar types of analysis to those presented above in relation to basketball have not been attempted for football.

OPEN QUESTION 7. *There has been significant research into producing spatially informed metrics for player and team performance in basketball; however, there has been little research in other sports, particularly football. It is an open research question whether similar spatially informed sports metrics could be developed for football.*

7. VISUALISATION

To communicate the information extracted from the spatio-temporal data, visualization tools are required. For real-time data, the most common approach is so-called *live covers*. This is usually a website that comprises a text panel that lists high-level updates of the key events in the game in almost real time and several graphics showing basic information about the teams and the game. Live covers are provided by leagues (e.g., National Hockey League, NBA, and Bundesliga), media (e.g. ESPN television network), and even football clubs (e.g., Liverpool and Paris Saint-Germain). For visualizing aggregated information, the most common approach is to use heat maps. Heat maps are simple to generate, are intuitive, and can be used to visualize various types of data. Typical examples in the literature are visualizing the spread and range of a shooter (basketball) in an attempt to discover the best shooters in the NBA [Goldsberry 2012] and visualizing the shot distance (ice hockey) using radial heat maps [Pileggi et al. 2012]. Two recent attempts to provide more extensive visual analytics systems have been made by Perin et al. [2013] and Janetzko et al. [2015].

Perin et al. [2013] developed a system for visual exploration of phases in football. The main interface is a timeline and *small multiples* providing an overview of the game. A *small multiple* is a group of similar graphs or charts designed to simplify comparisons between them. The interface also allows the user to select and further examine the *phases* of the game. A phase is a sequence of actions by one team bounded by the actions in which they first win and then finally lose possession. A selected phase can be displayed and the information regarding a phase is aggregated into a sequence of views, where each view only focus on a specific action (e.g., a long ball or a corner). The views are then connected to show a whole phase, using various visualization techniques, such as a passing network, a time line, and sidebars for various detailed information.

Janetzko et al. [2014, 2015] present a visual analysis system for interactive recogini-tion of football patterns and situations. Their system tightly couples the visualization system with data-mining techniques. The system includes a range of visualization techniques (e.g., parallel coordinates and scalable bar charts) to show the ranking of features over time and plots the change of game play situations, attempting to support the analyst to interpret complex game situations. Using classifiers, they automatically detected the most common situations and introduced semantically meaningful features for these. The exploration system also allows the user to specify features for a specific situations and then perform a similarity search for similar situations.

Open Question 8. *The area of visual interfaces to support team sports analytics is a developing area of research. Two crucial gaps are large user studies with the aim to explore (1) the analytical questions that experts need support for and (2) which types of visual analytical techniques can be understood by experts.*

8. APPLICABILITY OF APPROACHES TO OTHER SPORTS

The techniques and tools described in this survey are often motivated by a particular sport. This begs the question as to how easily the techniques could be applied to other team-based invasion sports. For the most part, our view is that the techniques are generally applicable to most other invasion sports; however, the peculiar rules and dynamics of each sport will impact on the specifics of the implementation. In this section, we provide a general discussion on some of the considerations inherent in applying techniques to various invasion sports.

Broadly speaking, the spatio-temporal analytic techniques described in this survey are designed to identify structure within spatio-temporal data, in particular, persistent formations and patterns. These are typically the manifestations of the tactical or strategic behaviour of the players, and such behaviour is common to all invasion sports. This objective, combined with other common features of all invasion sports, suggests that the techniques can be broadly applied. Examples of the common features are two opposing teams alternating between defence and attack, constrained playing area and match duration, the concept of possession of the ball/puck as a means of scoring a goal, and the physical and physiological constraints of nature and the players.

On the other hand, there are qualitative differences between various invasion sports that can impact how these techniques can be applied. Perhaps the most significant difference between sports in this context is the temporal segmentation of the match into *plays* that occur between stoppages and the frequency and variety of the events that occur within each *play*. At one extreme are sports such as American football, where each *play* is short, a single team is typically in possession for the duration of the *play*, and a small number of events can occur, such as passes or tackles. This can be contrasted with sports such as football, where a *play* may continue for an arbitrarily long time, possession may alternate between the two teams multiple times, and multiple events can occur. However, in both American football and football, a *play* will often end without a scoring opportunity, which is in contrast to basketball, where scoring events are more frequent.

These qualitative differences between *plays* impacts many of the analytic techniques described herein. For example, in sports like football, hockey, and rugby, where the possession of the ball can alternate multiple times within a *play*, spatial dominance of the entire playing area, described in Section 3, is perhaps more significant than a sport such as basketball, where the defending team will often cede the *back-court* and only seek to dominate the half-court below the basket they are defending or even just the area around the ball-carrier. Similarly, the network techniques described in Section 4 rely on sequences of passing events within a *play* and will be less applicable to sports where such sequences do not exist, like American football.

The research surveyed in Section 5 typically applies standard data-mining techniques to problems in the sporting domain and uses spatio-temporal data as the primary input data. The objectives of these works, such as event classification and prediction, formation and group movement identification, and temporal segmentation would be applicable to any invasion sport. This appears to be an effective strategy, and we expect continued interest in this approach.

A final consideration is the frequency in which particular events occur. Many of the techniques are predicated on the occurrence of events, most notably scoring events, and the number of times such events can occur varies among different sports. Thus, a technique that applies to basketball—where 50 or more scoring events occur per match—may be applicable in theory to football—where 2 or 3 scoring events per match is normal—however, in practice, a significantly larger number of matches must be available to effectively apply the technique.

9. CONCLUSION

The proliferation of optical and device tracking systems in the stadia of teams in professional leagues in recent years have produced a large volume of player and ball trajectory data, and this has subsequently enabled research efforts across a number of communities. To date, a diversity of techniques have been brought to bear on a number of problems described in this survey; however, there is little consensus on the key research questions or the techniques to use to address them. Thus, we believe that this survey of the current research questions and techniques is a timely contribution to the field.

This article surveys the recent research into team sports analysis that is primarily based on spatio-temporal data and describes a framework in which the various research problems can be categorised. We believe that the structured approach used in this survey reflects a useful classification for the research questions in this area. Moreover, the survey would be of value as a concise introduction for interested researchers who are new to the field.

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